

Project: Air Traffic Passenger Analytics

Tool: Microsoft Excel

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Project Overview

This project analyses Air Traffic Passenger Statistics data to uncover passenger traffic trends, airline performance, regional distribution, terminal utilization, and operational insights. The analysis was conducted using Microsoft Excel, leveraging data cleaning, transformation, Pivot Tables, Pivot Charts, KPIs, and an interactive dashboard to support data-driven decision-making.

The dataset contains passenger traffic records spanning from 1999 to 2025, covering over 1.1 billion passenger movements across multiple airlines, geographic regions, terminals, and boarding areas.

Data Cleaning and Preparation

Prior to analysis, the dataset was assessed for quality and completeness.

Data Cleaning Summary

- Checked all columns for missing values.
- Found 316 missing values in the Operating Airline IATA Code column; records were retained because the corresponding airline names were available.
- Identified 2 records with Passenger Count equal to 0, these were retained as valid observations.
- Verified that all remaining fields contained no significant missing values.
- Removed unnecessary metadata columns that were not required for analysis.
- Created helper columns including Year, Month, Year-Month, and Quarter to support trend analysis and dashboard filtering.

The dataset was subsequently deemed suitable for analysis.

Dashboard Analysis and Findings

Monthly Passenger Traffic Trend

Analysis of monthly passenger traffic revealed a strong long-term growth trend between 1999 and 2019. Passenger volumes steadily increased, surpassing 5 million passengers per month during peak periods.

A significant decline occurred in 2020 as a result of the COVID-19 pandemic, causing passenger traffic to fall to historic lows. However, beginning in 2021, passenger volumes experienced a strong recovery. By 2025, traffic levels had nearly returned to pre-pandemic levels.

Key Insight

Passenger traffic demonstrated resilience over time, recovering strongly following the pandemic and reaffirming sustained demand for air travel.

Top 10 Airlines by Passenger Volume

The airline analysis identified United Airlines as the dominant carrier throughout the study period.

Key Findings

- United Airlines transported more than 260 million passengers.
- Historical records categorized as "United Airlines - Pre 07/01/2013" contributed nearly 200 million additional passengers.
- American Airlines, SkyWest Airlines, and Delta Air Lines ranked among the next highest-volume carriers.
- A relatively small number of airlines accounted for a substantial proportion of total passenger traffic.

Business Insight

United Airlines maintained a dominant market position throughout the study period, highlighting its importance within the airport's operational network.

Passenger Traffic by GEO Region

Regional analysis revealed that domestic passenger traffic accounted for the majority of passenger movements.

Key Findings

- US passenger traffic represented approximately 76% of total passenger volume.
- Asia accounted for approximately 10% of passenger traffic.
- Europe represented approximately 7% of passenger traffic.
- Canada and Mexico contributed smaller but significant passenger volumes.

Business Insight

The airport primarily serves domestic travel demand while also functioning as a major international gateway, particularly for Asian and European markets.

Passenger Traffic by Terminal

Terminal utilization analysis identified significant differences in passenger volumes across airport terminals.

Key Findings

Terminal 3

- Processed approximately 454 million passengers.
- Accounted for roughly 41% of total passenger traffic.
- Recorded the highest passenger volume among all terminals.

International Terminal

- Processed over 282 million passengers.
- Served as a major hub for international travel.

Terminal 1

- Processed more than 261 million passengers.
- Maintained consistently high passenger volumes.

Terminal 2

- Processed approximately 111 million passengers.
- Recorded the lowest traffic among major terminals.

Business Insight

Terminal 3 serves as the airport's primary operational hub, while the International Terminal reinforces the airport's role as a key international gateway.

Passenger Traffic by Activity Type

Passenger activity analysis was conducted to understand traffic distribution across different passenger movement categories.

The results highlighted the balance between arriving, departing, and transit passengers, providing insight into overall airport operational activity and passenger flow patterns.

GEO Summary Analysis

The GEO Summary analysis compared domestic and international traffic volumes.

Key Findings

- Domestic traffic accounted for the majority of passenger movements.
- International traffic represented a substantial portion of total passenger volume.

Business Insight

The airport functions as both a major domestic transportation hub and an important international gateway connecting passengers to global destinations.

Boarding Area Analysis

Boarding area analysis provided additional operational insights regarding passenger concentration within the airport.

Key Findings

Boarding Area F

- Processed approximately 335 million passengers.
- Recorded the highest passenger volume among all boarding areas.

Boarding Area B

- Processed approximately 149 million passengers.

Boarding Area G

- Processed approximately 145 million passengers.

Other Boarding Areas

- Boarding Areas C, D, and E handled substantially lower passenger volumes.

Business Insight

Passenger traffic is concentrated within a small number of boarding areas, with Boarding Area F serving as the airport's busiest operational zone.

Passenger Traffic by Price Category

Passenger traffic was analyzed by fare category to understand the contribution of low-fare carriers.

Key Findings

- Non-low-fare carriers accounted for approximately 87% of total passenger traffic.
- Low-fare carriers represented approximately 13% of passenger volume.

Business Insight

Traditional carriers continue to dominate passenger traffic, while low-fare carriers contribute a smaller but meaningful share of overall airport activity.

Conclusion

This analysis of over 1.1 billion passenger movements provided valuable insights into airport operations, airline performance, regional travel demand, terminal utilization, and passenger behavior.

The findings revealed sustained long-term growth in passenger traffic, significant disruption during the COVID-19 pandemic, and a strong recovery in subsequent years. United Airlines emerged as the dominant carrier, Terminal 3 was identified as the busiest terminal, and Boarding Area F processed the highest passenger volume. Domestic travel accounted for the majority of traffic, while Asia represented the largest international market.

The interactive dashboard developed in Microsoft Excel enables stakeholders to monitor key performance indicators, explore passenger trends, and gain actionable insights into airport operations.

Tools Used

- Microsoft Excel

Skills Demonstrated

- Data Cleaning
- Data Transformation
- Exploratory Data Analysis (EDA)
- Data Visualization

- KPI Development
- Dashboard Design
- Business Intelligence Reporting